

Project Plan

Ku CampusBite Food Delivery System

Prepared For:

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Course Title : Software Project Management

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Revision History

Version	Date	Author(s)	Summary of Changes
1.0	July 30, 2025	Cantech Team	Initial draft and baseline project plan.
1.1	Aug 16, 2025	Sana GhulamRasool (Project Manager)	Introduction: New section added. Project Objectives: Updated to a SMART format. Scope: Replaced "Canteen Dashboard" with "WhatsApp API Integration" to simplify order notifications for canteen staff. Also, a new "Optional Features" section was added. WBS: Revised for greater detail and accuracy.
1.2	Sep 25, 2025	Sana GhulamRasool (Project Manager)	Project Schedule: Updated based on the detailed WBS from Version 1.1 and now includes a Project Schedule Summary and a schedule derived from a network diagram. Roles and Resources: Updated to reflect necessary reallocation of tasks following the temporary absence of a core team member.
1.3	Oct 10, 2025	Sana GhulamRasool (Project Manager)	Scope: The feature Canteen Order Notification was formally reduced to an Automated Notification Proof-of-Concept (POC) via Change Request, due to budget constraints on the paid API service. Risk Identification & Mitigation: Risks were re-assessed using a Probability/Impact Grid.

Introduction

The KU CampusBite Food Delivery System is a software project designed to streamline the food ordering and delivery process at a large university campus. This revised project plan (Version 1.1) outlines the refined scope, updated objectives, and key deliverables for the system, which will be developed to ensure efficient and timely service for both students and faculty.

Project Objectives

1. **Project Duration:** 3 months
2. **Reduce Student Travel Costs** – Achieve up to **44% cost savings per order** by replacing physical travel to distant canteens (Rs. 90 average auto fare) with platform-based delivery (Rs. 50 or free).
3. **Save Student Time** – Ensure food delivery within **30–35 minutes** (including preparation + delivery), reducing the current **45–60 minutes of commuting and queuing**.
4. **Increase Accessibility & Reach** – Provide a **centralized ordering platform for all 15 canteens**, enabling students easy access while helping canteens **expand customer reach by 10–20%**.
5. **Enhance Efficiency** – Improve order processing speed by **20% through automation**, reducing manual errors and waiting times.
6. **Ensure Student Satisfaction** – Achieve **60–65% positive feedback** on usability, convenience, and reliability within the **first 3 months** of implementation.

Project Scope

Included:

- 1 Features for meal pre-ordering, order-scheduling from multiple KU canteens.
- 2 Secure user and admin login, plus guest access without account creation.
- 3 Student facing interface for browsing menus, searching items, managing carts, and placing orders.
- 4 Order details automatically sent to the respective canteen via Whatsapp.
- 5 Mobile-responsive website design for easy access from smartphones and laptops.
- 6 Onboarding of canteen staff with guidance on using the system and updating item availability status.

Excluded:

- 1 Payment gateway integration - initial version will use cash on delivery.
- 2 Integration with student ID cards or university academic systems.
- 3 Off-campus deliveries – service is strictly for Karachi University campus.
- 4 Multilingual interface – initial version will be in English only.
- 5 Native mobile application – the system will be web-based only.
- 6 Real-time delivery tracking on a map.

Optional Features

- 1 Menu filtering – price-based and canteen-based filtering options
- 2 Ratings and feedback – user reviews and ratings for canteens and orders

- 3 Menu availability update tool – online tool for canteen staff to update item availability
- 4 Help/FAQ section – quick-start guide for students on using the website

Work Breakdown Structure (WBS)

1. Project Initiation & Documentation

- 1.1 Define project objectives and goals
- 1.2 Identify key stakeholders and roles
- 1.3 Prepare initial project charter and scope outline
- 1.4 Documentation (maintaining all key project documents and updating them as needed throughout lifecycle)

2. Requirements Gathering & Analysis

- 2.1 Gather requirements from stakeholders
- 2.2 Identify functional requirements
- 2.3 Identify non-functional requirements (usability, availability, etc.)

3. Design

- 3.1 UI/UX Design (basic interface structure, initial layout decisions)
- 3.2 Database Design (ER diagram, schema notes)
- 3.3 Use Case Diagram (system interaction representation)

4. System Development

- 4.1 Student Interface Development
- 4.2 Login & Authentication
- 4.3 Admin Dashboard Development
- 4.4 Backend & Frontend Integration
- 4.5 WhatsApp Integration for canteen notifications

5. Testing & Quality Assurance

- 5.1 Unit Testing (testing each module individually)
- 5.2 Integration Testing (testing how modules work together — e.g., login + student interface + database)
- 5.3 System Testing (end-to-end testing of the entire application)
- 5.4 User Acceptance Testing (stakeholders test if the system meets needs)
- 5.5 Test additional features after implementation

6. Enhancements & Improvements

- 6.1 Implement additional features (if time/resources allow)
- 6.2 Improve UI based on testing and feedback
- 6.3 Re-test enhancements

7. Deployment & Launch

7.1 Prepare deployment environment

7.2 Deploy final application

7.3 Official project launch

Schedule

Project Schedule Summary

- **Project Start Date:** July 3, 2025
- **Planned Completion Date:** September 22, 2025
- **Official Deadline :** October 14, 2025

Detailed Project Schedule

Task ID	Task	Start Date	End Date	Duration (Days)	Person Responsible
1.0 Planning & Scoping					
1.1	Define Project Objectives and Goals	03-Jul	04-Jul	2	Project Manager
1.2	Identify Key Stakeholders and Roles	03-Jul	03-Jul	1	Project Manager
1.3	Prepare Initial Project Charter & Scope	05-Jul	07-Jul	3	Project Manager
1.4	Create Initial Project Plan (Schedule, Budget, Risks)	08-Jul	11-Jul	4	Project Manager
2.0 Requirements Analysis					
2.1	Gather Requirements from Stakeholders	12-Jul	16-Jul	5	Business Analyst
2.2	Identify Functional Requirements	17-Jul	19-Jul	3	Business Analyst
2.3	Identify Non-Functional Requirements	17-Jul	19-Jul	3	Business Analyst
3.0 Design					
3.1	UI/UX Design & Mockups	20-Jul	26-Jul	7	UI/UX Designer
3.2	Database Design (ERD, Schema)	20-Jul	24-Jul	5	Database Engineer
3.3	System Architecture & Use Case Diagram	20-Jul	23-Jul	4	System Architect

4.0 Development					
4.1	Student Interface Development	27-Jul	10-Aug	15	Frontend Developer
4.2	Login & Authentication Module	27-Jul	31-Jul	5	Backend Developer
4.3	Admin Dashboard Development	27-Jul	05-Aug	10	Full Stack Developer
4.4	Backend & Frontend Integration	11-Aug	22-Aug	12	Full Stack Developer
4.5	WhatsApp Integration	23-Aug	27-Aug	5	Integration Specialist
5.0 Testing & QA					
5.1	Unit Testing (Code Level)	28-Aug	30-Aug	3	QA Engineer
5.2	Integration Testing (Modules)	31-Aug	03-Sep	4	QA Engineer
5.3	System Testing (End-to-End)	04-Sep	08-Sep	5	QA Lead
5.4	User Acceptance Testing (UAT)	09-Sep	11-Sep	3	Client + QA
6.0 Enhancements & Re-testing					
6.1	Implement Additional Features	12-Sep	17-Sep	6	Development Team
6.2	Improve UI Based on Feedback	12-Sep	15-Sep	4	UI/UX Designer
6.3	Re-test Enhancements	18-Sep	20-Sep	3	QA Engineer
7.0 Deployment & Launch					
7.1	Prepare Deployment Environment	12-Sep	14-Sep	3	DevOps Engineer
7.2	Deploy Final Application	21-Sep	21-Sep	1	DevOps Engineer
7.3	Official Project Launch	22-Sep	22-Sep	1	Project Manager

Milestones:

Milestone	Target Date
Requirements Finalization	31 July 2025
UI Design & Core Backend Structure Implemented	18 Aug 2025
Development Completed	15 Sept 2025
Integrated Testing	24 Sept 2025
Final Deployment	8 Oct 2025

Roles and Resources

Role/Resource	Person/Team
Project Manager	Sana Ghulam Rasool
Business Analyst	Sania Khan
UI/UX Design Team	Sania Khan, Humaima Moin, Lamia Gull
Frontend Development Team	Sania Khan, Humaima Moin, Lamia Gull, Neha Rehan
Backend Development Team	Sana Ghulam Rasool
Quality Assurance Team	Sana Ghulam Rasool, Neha Rehan
End User Representatives	Classmates (Students)
DevOps Team	Cantech Team

Risk Identification and Mitigation

RISK ID	Risk Name	Risk Description	Priority (H/M/L)	Mitigation Strategy
1	API Integration	WhatsApp API fails to connect, stopping order notifications to canteens	High	Build offline order support with auto-sync and use SMS fallback for critical updates.
2	Unrealistic Deadlines	Ambitious deadlines lead to rushed work and lower-quality code.	High	Set realistic timelines with canteen partners and developers to ensure proper testing and stable releases
3	Communication Failure	Lack of team communication causes confusion and duplicated work.	High	Hold regular meetings between canteen owners and developers to ensure smooth coordination.
4	Scalability	Website crashes during peak hours due to high traffic volume.	Medium	Build the system with scalable architecture to handle increasing student traffic and canteen orders smoothly.

5	Team Member Absence	A key team member becomes unavailable, halting progress and creating a knowledge gap.	High	Reduce dependency on individuals by maintaining shared documentation, and reassigning tasks when a member is unavailable.
6	Buggy Code	The final code contains multiple bugs, leading to order errors and a poor user experience.	Medium	Implement testing at each stage, review code regularly, and use a bug-tracking system to find and fix errors early.
7	Lack of Expertise	The team lacks skills for advanced tasks, leading to development delays.	High	Enhance team skills through peer mentoring, online tutorials, and supervisor guidance to ensure all modules (backend, frontend, database) are completed efficiently.
8	Financial Constraints	Limited budget prevents the purchase of necessary tool , APIs or reliable hosting.	Medium	Minimize costs by using free-tier hosting services, open-source tools and APIs and budgeting resources carefully to stay within financial Limits.

Communication Plan

Purpose	Who will they talk	How they will talk	When
Share the project progress	Cantech team	Weekly meeting	Every Monday
Design and Feature discussions	Developer	WhatsApp or a meeting	2 times a week or as needed
Ask for feedback	Stakeholders	Survey	During the testing phase
Report problems or bugs	Developers	WhatsApp	As soon as the issue happens
Final update and project delivery	Cantech and developers	Presentation or meeting	At the end of the project

Change Control Process

Steps	What will happen
Request Change	A team member, student, or supervisor suggests a change
Record the Change	Team writes it down in a change log
Review the Change	Team checks that it is useful, can we do it in time? Does it affect other parts?
Approve or Reject	PM and team decide to approve or reject
Make the Change	A developer or a designer makes the change in the system
Inform everyone	Team informs all members and updates documentation